MILITARY SPECIFICATION

WORK BOX, DENTAL LABORATORY

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of work box suitable for holding dental prosthetic cases.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specification and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Set-Up.

Military

MIL-P-116 - Preservation, Methods of.
MIL-L-10547 - Liners, Case and Sheet; Overwrap; Water-Vapor-proof or Waterproof, Flexible.

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Material, Code RSTS, 2800 South 20th Street, Philadelphia, PA 19101-8419, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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STANDARDS

Federal

FED-STD-66 - Steel, Chemical Composition and Hardenability.
FED-STD-151 - Metals, Test Methods.

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

DRAWING

Defense Personnel Support Center (DPSC)

10122 - Work Box, Dental Laboratory

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

3. REQUIREMENTS

3.1 Material.

3.1.1 Work box. The work box shall be fabricated from type 302, 304 or 430 corrosion-resisting steel of FED-STD-66.

3.1.2 Card shield. The card shield shall be fabricated from clear methyl methacrylate.

3.2 Style, design and dimensions. The steel work box shall be suitable for holding dental prosthetic cases. The clear plastic card shield shall be suitable for viewing patient identification cards. The style, design and dimensions of the work box shall conform to Defense Personnel Support Center (DPSC) Drawing No. 10122.

3.3 Drawings. The drawings forming a part of this specification are not production drawings. Drawings establish dimensional limitations and control design and configuration. The supplier shall be responsible for preparing his own shop drawings. The supplier shall select dimensions and tolerances within the limitations prescribed on the drawings to provide a product with proper design, configuration, fit and function. No deviations from the dimensional limitations prescribed on the drawings are permissible without prior approval of the contracting officer.

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3.4 Construction. Boxes shall be constructed to permit stacking, without binding or excessive play, when tested as specified in 4.4.2. One plastic card shield shall be furnished with each work box, and shall fit in outer and inner channels of box without binding.

3.5 Welding. The welding shall be durable and sound in every detail. Component members shall be carefully and properly aligned prior to welding into permanent position as a straight, rigid assembly. All welding shall be done with appropriate equipment properly regulated and operated by thoroughly trained and experienced welders highly skilled in the operations of stainless steel welding.

3.6 Finish.

3.6.1 Box. All exposed outside surfaces of the boxes shall have a commercial No. 2B or better finish for work boxes fabricated from sheet stock, and a No. 2 or better finish for work boxes fabricated from strip stock. Interior surfaces shall have a uniform mill finish. Boxes shall be thoroughly cleaned to remove foreign matter, such as dirt, grease, and metal. Exposed spot welds shall be ground smooth. Spot welds that are smooth need not be ground down.

3.6.2 Card shield. The plastic card shield shall have a smooth finish.

3.7 Identification marking. Each box shall be permanently and legibly marked on the bottom outside with the name or registered trade-mark of the manufacturer.

3.8 Workmanship. The work box and card shield shall be free from defects which detract from their appearance or impair their serviceability.

3.9 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Governments's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel, and plastic materials, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component or material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements, the lot number, the lot size and the sample size .

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables I, II and III (see 4.2.1, 4.2.2 and 4.3).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in table I.. The unit of product for examination purposes shall be one box.

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TABLE I. Examination criteria.

Type defect	Inspection Level	AQL (Percent Defective)
Major	S-2	1.0
Minor	S-2	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table II. The unit of product for test purposes shall be one box.

TABLE II. Sampling criteria for tests.

Characteristic	Requirement	Test procedure	Inspection level	AQL
Chemical Analysis	3.1.1	4.4.1	See sampling of raw materials	*
Stacking	3.4	4.4.2	S-2	1.0

* Acceptance number shall be zero for all sample sizes.

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in the construction of the box shall be selected for chemical analysis.

4.3 Examination. The boxes shall be examined for defects including, but not limited to those listed in table III.

TABLE III. Classification of defects.

Categories	Defects
Critical	None defined.
Major:	
101	Style, design and construction of work box and card shield not as specified.
102	Exposed outside surface of work box not commercial finish as specified.
103	Work box not free from sharp edges.
104	Box rocks on flat surface.

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TABLE III. Classification of defects (continued).

Minor:	
201	Surface of work box not free from rough grind marks, pits, scales, and dents.
202	Work box not thoroughly clean.
203	Identification marking not permanently and legibly marked.
204	Exposed spot welds not smooth.
205	Plastic card shield not clear.
206	Plastic card shield not smooth finish.
207	Dimensions of work box and card shield not within tolerances specified.
208	Card shield does not fit in channels as specified.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Chemical analysis. The chemical analysis shall be conducted in accordance with methods 111 or 112 of FED-STD-151. In the event of conflict, method 111 shall govern.

4.4.2 Stacking. The work boxes shall be stacked. The boxes shall be rejected when they do not nest, when nested boxes bind, and when a nested box has more than 3/16-inch horizontal movement.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Work box shall be cleaned and dried in accordance with MIL-P-116.

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5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. Each work box shall be packaged in accordance with MIL-P-116, Method IC-1.

5.1.2 Level C.

5.1.2.1 Unit package. Preservation for work box shall conform to the requirements of MIL-STD-794 for level C.

5.2. Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Twenty-four unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2, or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specifid in PPP-B-636.

5.2.2 Level B. Twenty-four unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1, or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for work box shall be packed in accordance with the requirements of MIL-STD-794 for level C.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination exceed 250 pounds or 20 cubic feet. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 20 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 48 inches in length, 52 inches in width, and 54 inches in height. Pallet shall conform to Type IV or Type V.

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5.2.5.1 Levels A and B. Work box packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. Work box packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6520-00-514-2394	WORK BOX, DENTAL LABORATORY

6.3 Subject term (key word) listing:

Work box
Prosthetic cases
Dental Laboratory

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MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Review Activities:

Army - MD
Navy - MS
Air Force - 03

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

47887

Preparing Activity:

DoD-MB

AGENT:

DLA-DM

Project No. 6520-2072

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-W-42059	WORK BOX, DENTAL LABORATORY

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR
b ADDRESS (Street, City, State, ZIP Code)	☐ USER
	☐ MANUFACTURER
	☐ OTHER (Specify)

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER (Last, First, MI) – Optional	b WORK TELEPHONE NUMBER (Include Area Code) – Optional
c MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8 DATE OF SUBMISSION (YYMMDD)

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